

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

Duration :60 Minutes

Code of Conduct

I M.Sandeep (192525095) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sandeep M

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
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|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |
- | | | | |
|--|--------------|----------------|----------------|
| Agriculture | Smart | Farming | Problem |
| A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated | | | |

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

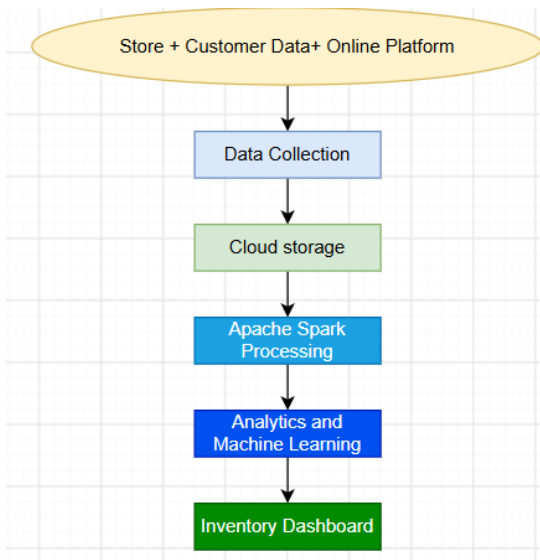
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem

A retail company receives data from physical stores, online shopping websites, warehouses, mobile applications, and customer loyalty programs. Since the data is generated continuously, a big data system is needed to collect, store, process, and analyze it.

Simple Architecture



Working

Sales from physical stores and online orders can be collected using Apache Kafka. Inventory updates, customer clicks, product searches, and purchase information are also sent to the system.

The collected data can be stored in a data lake using Hadoop HDFS or cloud storage. Apache Spark can process both historical and current data.

For real-time inventory tracking, every sale automatically reduces the available product quantity. If the stock level becomes low, the system can generate an alert for the warehouse or store manager.

For demand forecasting, historical sales data, seasonal trends, festivals, promotions, and customer demand can be analyzed. Models such as Linear Regression, Random Forest, XGBoost, and LSTM can be used to predict future product demand.

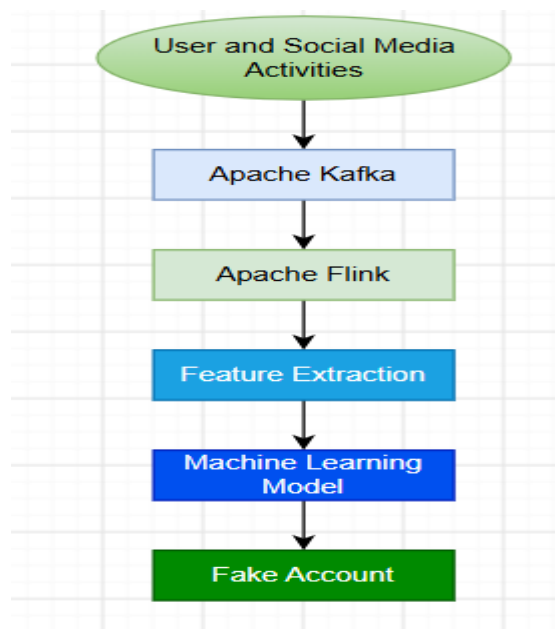
For personalized marketing, a recommendation system can analyze previous purchases and browsing behavior. Collaborative Filtering can recommend products that are popular among customers with similar interests.

The final results can be displayed through Power BI or Tableau dashboards. This system helps the company avoid stock shortages, improve sales planning, and provide better product recommendations.

2. Social Media Fraud Detection Problem

A social media platform receives a very large number of posts, comments, likes, shares, and account activities every second. The main challenge is identifying fake accounts and misinformation as quickly as possible.

Simple Architecture



Working

All user activities can be collected using Apache Kafka. Kafka can handle a large number of messages and distribute them to different processing systems.

A stream processing framework such as Apache Flink or Spark Streaming can process the incoming data continuously. The system can examine user behavior such as:

- Number of posts in a short time
- Repeated comments
- Number of followers
- Account creation date

- Unusual login activity

Machine learning models such as Random Forest, XGBoost, and Logistic Regression can classify accounts as genuine or suspicious.

For unknown or unusual behavior, Isolation Forest or anomaly detection techniques can be used.

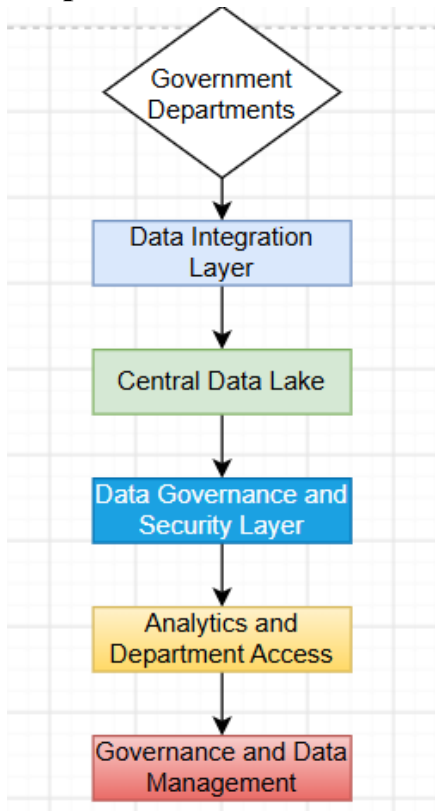
For misinformation detection, Natural Language Processing (NLP) can analyze the text in posts and comments. Models such as BERT or other deep learning models can classify content as potentially misleading or suspicious.

Suspicious accounts can be temporarily restricted or sent for manual verification. This architecture supports real-time detection and can scale by adding more Kafka and processing nodes.

3. Government Data Platform Problem

A government data platform needs to combine information from different departments. The data may include census records, economic data, geographical information, education records, and other public datasets.

Simple Architecture



The first step is to establish common data standards. Different departments may store information in different formats, so standard schemas and formats should be defined.

A central data lake can store large volumes of structured and unstructured data. Technologies such as Hadoop, cloud storage, and distributed databases can be used.

A data catalog should maintain information about available datasets, data owners, formats, and sources. Tools such as Apache Atlas can help manage metadata and data lineage.

For interoperability, government departments can use APIs and common data formats such as JSON, CSV, XML, and standardized schemas.

Security can be managed using:

- Role-Based Access Control
- User authentication
- Data encryption
- Data masking
- Tokenization
- Audit logs

Sensitive personal information should only be available to authorized users. For public analysis, personal data can be anonymized.

Apache Ranger or similar security tools can be used to manage access permissions.

This approach provides a common platform where departments can access reliable data while maintaining privacy, security, and interoperability.

4. Banking Fraud Detection Problem

A bank processes a large number of transactions from ATMs, credit cards, mobile banking, online banking, and money transfers. Fraudulent transactions must be detected quickly because even a small delay can result in financial loss.

The system should process both historical transaction data and live transaction streams.

Simple Architecture

ATM + Card + Mobile + Online Transactions

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Apache Kafka



Apache Flink / Spark Streaming



Feature Extraction and Fraud Analysis



Machine Learning Model



Approve / Block / Alert

Working

Every transaction is sent to Apache Kafka. Kafka distributes transactions to multiple processing systems.

Apache Flink or Spark Streaming processes the transactions in real time. The system can analyze information such as:

- Transaction amount
- Location
- Time
- Device used
- IP address
- Merchant type

Historical data can be stored in Hadoop HDFS, cloud storage, or a data lake. Previous transactions help the system understand normal customer behavior.

Machine learning algorithms can include:

- Logistic Regression for basic classification
- Random Forest for detecting known fraud patterns
- XGBoost for high-performance classification
- Isolation Forest for unusual transactions
- Autoencoders for anomaly detection

A fraud score can be generated for every transaction.

For example:

- Low score: Transaction is approved.
- Medium score: Additional verification may be required.
- High score: Transaction is blocked or immediately flagged.

A combination of rule-based detection and machine learning can improve accuracy. This architecture provides fast processing and helps reduce financial fraud.

5. Agriculture Smart Farming Problem

A smart farming system receives data from soil sensors, weather stations, drones, cameras, and agricultural equipment. These devices generate data continuously from different locations.

The purpose of the system is to help farmers make better decisions about irrigation, fertilizers, crop health, and resource usage.

Simple Architecture

Soil Sensors + Weather Stations + Drones

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IoT Gateway / MQTT

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Apache Kafka

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Data Lake / Cloud Storage

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Apache Spark Analytics

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Machine Learning Models

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Farmer Mobile App / Dashboard

Working

Soil sensors can collect information such as:

- Soil moisture
- Temperature

- Soil pH
- Humidity

Weather stations can provide rainfall, temperature, and weather conditions. Drones can capture images of crops and agricultural fields.

IoT devices can send data through MQTT, while Apache Kafka can collect large volumes of data from different sources.

The data can be stored in a data lake using Hadoop or cloud storage. Apache Spark can analyze large amounts of historical and real-time agricultural data.

Machine learning can support farmers in different ways.

Crop Yield Prediction

Models such as Linear Regression, Random Forest, and XGBoost can predict crop production using previous crop, soil, and weather data.

Smart Irrigation

Soil moisture sensors and weather forecasts can determine when irrigation is required. The system can recommend the required amount of water.

Crop Disease Detection

Images captured by drones can be analyzed using Convolutional Neural Networks (CNN). The system can identify possible crop diseases or unhealthy areas.

Fertilizer Recommendation

The system can analyze soil conditions and recommend the correct amount of fertilizer.

Decision Support

The final information can be shown through a mobile application or dashboard. Farmers can receive alerts such as:

- Low soil moisture
- Possible crop disease
- Heavy rainfall expected
- High temperature warning

- Irrigation required

This big data solution supports precision agriculture. It helps farmers improve crop yield, reduce water and fertilizer wastage, and make decisions based on real-time information.